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4 authors, including:

David Poole
Kansas State University

Susan A. Ward
Human Bio-Energetics Research Centre, Crickhowell, Powys

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Metabolic and respiratory profile of the upper limit for prolonged exercise in man

DAVID C. POOLE, SUSAN A. WARD†, GERALD W. GARDNER and BRIAN J. WHIPP

Departments of Anesthesiology and Kinesiology, UCLA, Los Angeles, California 90024, U.S.A.

and

Division of Respiratory Physiology and Medicine, Department of Medicine, Harbor-UCLA Medical Center, Torrance, California 90509, U.S.A.

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For high-intensity cycling, power (P) can be well described as a hyperbolic function of tolerable work duration (t): $P=(W'/t)+P_{LL}$. W′ is a constant and P_{LL} is the lower limit (asymptote) for P which is shown to occur at an O_2 uptake ($\dot{V}O_2$) lying above the estimated threshold for sustained blood [lactate] increase ($\hat{\Theta}_{lac}$) but below the maximum $\dot{V}O_2$ ($\dot{V}O_2$max) obtained during incremental cycling. This relation suggests that, above P_{LL}, only a certain *amount* of work (W′) can be accomplished regardless of its rate of performance, with $\dot{V}O_2$max being attained at fatigue. Hence, P_{LL} defines a point of discontinuity in the $\dot{V}O_2$ –P relation for supra-Θ_{lac} exercise. In order to determine the factors responsible for the continued increase in $\dot{V}O_2$ (to the maximum fatiguing value) at power outputs $>P_{LL}$, we documented the temporal profiles of metabolic (rectal temperature; blood [lactate], [pyruvate], [norepinephrine], [epinephrine]) and respiratory ($\dot{V}_E$; $\dot{V}O_2$; $\dot{V}CO_2$; blood pH, PCO_2, $[HCO_3^-]$) responses to constant-load cycling in eight healthy males at P_{LL} (24 min) and slightly above P_{LL} (to exhaustion, i.e. <24 min). $\dot{V}O_2$ manifested a delayed steady state at P_{LL}, despite catecholamine levels and core temperature continuing to increase throughout; blood [lactate] and pH plateaued, however. In contrast, $\dot{V}O_2$ continued to increase slowly for the duration of the exercise $>P_{LL}$ and attained $\dot{V}O_2$max. The response patterns at P_{LL} and $>P_{LL}$ suggest that the slow phase of the $\dot{V}O_2$ response is best correlated with the temporal profile of blood [lactate], and hence the site and route of metabolism of this variable may play a major role in the $\dot{V}O_2$ kinetics for high-intensity exercise.

1. Introduction

The duration (t) for which high-intensity dynamic exercise can be performed to the point of fatigue is inversely proportional to the power (P) (Hill 1927, Monod and Scherrer 1965, Moritani *et al.* 1981, Hughson *et al.* 1984). This P–t relation is well described by a hyperbolic function: $(P-P_{LL}).\,t=W'$ (eqn. 1), where W′ represents a constant amount of work that normally can be performed *above* the lower limit or asymptote of power (P_{LL})‡, regardless of the rate at which it is performed. This relation may be transformed into its linear formulation: $P=(W'/t)+P_{LL}$ (eqn. 2),

† To whom correspondence should be addressed at Department of Anesthesiology, C.H.S. UCLA, Los Angeles, California 90024, U.S.A., Tel. (213) 206-6241.

‡ The parameter P_{LL} is the power asymptote for the power–duration relationship and thus represents a unique power output above which the time to fatigue is predictably determined for a hyperbolic relationship.

where W′ and P_{LL} are the slope and intercept, respectively. It is important to emphasize that this relation is unlikely to provide a precise representation of the actual physiological behaviour at the very extremes of performance, as distorting factors such as (i) limitations of mechanical force-generation for the highest power requirements (corresponding to a very short exercise duration) and (ii) substrate provision, thermoregulatory or body-fluid requirements for markedly prolonged exercise provide further constraint.

The parameter P_{LL}, which has been shown to lie above what has been variously termed the lactate threshold (Θ_{lac}), the anaerobic threshold and the ventilatory threshold‡, represents an asymptotic or 'critical' power (Moritani *et al.* 1981). Hence, for power falling between Θ_{lac} and P_{LL}, there is presumably the capability of attaining a steady state of O_2 uptake ($\dot{V}O_2$) and thus of performing prolonged exercise. Beyond P_{LL}, however, work performed to exhaustion is predictably of limited duration, and for which the end-point $\dot{V}O_2$ approaches (or may even exceed) the maximum ($\dot{V}O_2$max) as estimated from conventional incremental testing. Consequently, P_{LL} can be regarded as an important functional demarcator of work intensity, with implications for the potential tolerable duration of a particular work task.

Several investigators have attempted to determine this power asymptote in man by establishing the parameters of the *P–t* relation (Monod and Scherrer 1965, Moritani *et al.* 1981, Wilkie 1981, Hughson *et al.* 1984). To date, however, there has been no empirically determined physiological characterization of P_{LL} or of the pattern of metabolic and respiratory change which leads to the inexorable fatigue at supra-P_{LL} power. We therefore investigated the temporal profiles of metabolic and respiratory responses to prolonged, constant-load cycle ergometry both at and slightly above P_{LL}.

2. Methods

Eight healthy, young males (table 1), none of whom was involved in regular physical training, performed incremental and square-wave exercise tests on an electromagnetically braked cycle ergometer (Mijnhardt, #KEM-2); pedalling frequency was maintained at ~60 min^{-1}. Expired flow was measured with a pneumotachograph (Rudolph, #3800) downstream of the expiratory port of a low-resistance breathing valve (Rudolph, #2700, dead space: 90 ml), connected to a variable-reluctance manometer (Validyne, #MP45: ± 2 cm H_2O); the pneumotachograph was maintained at 37°C by a thermal feedback device and calibrated with known volumes of room air at several mean flows. Rapidly responding analysers (calibrated with precision-analysed gases) continuously monitored PCO_2 (Datex, #CD 101; 90% response time: 0·1 s) and PO_2 (Applied Electrochemistry, #S3A: 90% response time: <0·08 s) in respired gas sampled from the mouthpiece. Heart rate was derived beat-by-beat from the R–R interval of an electrocardiograph signal (Birtcher, #7000). The electrical signals from these devices underwent analog-to-digital conversion and computer analysis (Tektronix, #4052) for on-line, breath-to-breath determination of ventilation ($\dot{V}_E$, BTPS), tidal volume ($\dot{V}_T$, BTPS), frequency (*f*), CO_2 output ($\dot{V}CO_2$, STPD), O_2 uptake ($\dot{V}O_2$, STPD), ventilatory equivalents for CO_2 and O_2 ($\dot{V}_E/\dot{V}CO_2$, $\dot{V}_E/\dot{V}O_2$), respiratory exchange ratio (*R*), and end-tidal PCO_2 and PO_2 ($P_{ET}CO_2$, $P_{ET}O_2$).

Each subject performed an incremental exercise test (25 W min^{-1}) to the limit of

‡ The terms '*anaerobic threshold*' and '*ventilatory threshold*' are used to represent the $\dot{V}O_2$ at which blood [lactate] begins to increase systematically during a particular form of exercise, as estimated by $P_{ET}O_2$ and $\dot{V}_E/\dot{V}O_2$ starting to increase without a concomitant increase in $\dot{V}_E/\dot{V}CO_2$ and a decrease in $P_{ET}CO_2$.

Table 1. Physical and exercise characteristics of subjects.

Subject	Age (yr)	Weight (kg)	Height (cm)	$\hat{\Theta}_{lac}$ (l min^{-1})	$\hat{\Theta}_{lac}$ (W)	$\dot{V}O_2$ max (l min^{-1})	$\dot{V}O_2$ max (ml min^{-1} kg^{-1})	P_{max} (W)
1	21	69·1	175·2	1·79	125	3·72	53·8	300
2	19	94·6	177·7	1·59	107	3·87	50·5	275
3	22	92·3	190·4	2·04	137	4·66	40·9	350
4	20	65·0	170·1	1·42	87	3·53	54·3	250
5	23	68·2	167·6	1·75	126	3·31	48·5	250
6	22	61·5	172·7	1·52	95	3·07	49·9	250
7	25	71·0	170·0	1·72	150	3·60	50·7	300
8	24	83·1	180·3	1·93	129	4·68	56·3	325
Mean	22	75·6	175·5	1·72	120	3·81	50·6	288
S.E.	1	4·8	7·4	0·08	8	0·22	1·7	14

$\hat{\Theta}_{lac}$, $\dot{V}O_2$max and P_{max} represent the lactate threshold (expressed in units of $\dot{V}O_2$ and power), the maximal $\dot{V}O_2$ (expressed in absolute units and normalized for body weight) and the maximal power output for the incremental exercise test.

tolerance for the estimation of Θ_{lac} and $\dot{V}O_2$max. Θ_{lac} was established as the $\dot{V}O_2$ at which $P_{ET}O_2$ and $\dot{V}_E/\dot{V}O_2$ started to rise systematically without a simultaneous rise in $\dot{V}_E/\dot{V}CO_2$ and fall in $P_{ET}CO_2$ (Whipp and Mahler 1980); it should be noted that $\hat{\Theta}_{lac}$ does not depend on the rate at which work rate is incremented when it is expressed as an O_2 uptake (Whipp *et al.* 1981). $\dot{V}O_2$max was established as the highest $\dot{V}O_2$ attained. To define the *P–t* relation for high-intensity exercise, each subject undertook a series of a least five different fatiguing square-wave tests from a 5-min base-line of unloaded pedalling, avoiding power outputs which were so great that they induced exhaustion in less than 1 min. The time to fatigue was taken as the interval between the imposition of the load and the point at which the subject could no longer continue cycling and therefore sustain the required power (this being independent of pedalling frequency for our ergometer). Only one test was undertaken on a given day and the sequence was randomized; the subjects were not told for how long or at what power they had exercised. Reproducibility was assessed by repeat tests assigned randomly to subjects. The time to fatigue at a given power for a given subject showed only minor variation: e.g. <5–10 s (<2–4%) for tests of ~ 4 min duration, and <20–30 s (<4–6%) for tests of ~ 8 min duration. The *P–t* relation was linearized (eqn 2); the slope (W′) and intercept (P_{LL}) parameters were extracted by least-squares linear regression techniques.

Each subject then performed two further square-wave tests (on different days) at P_{LL} and $P_{LL}+5\%$ of P_{max} on the incremental test ($P_{LL}+15$ W, on average). The work was continued for 24 min or until the subject could no longer sustain the power. Rectal temperature was monitored (Yellow Springs Instrument Co., #400/44TA). Arterialized venous blood was sampled from an in-dwelling catheter (18 ga.) in a superficial vein on the heated dorsum of the hand (Forster *et al.* 1972). Samples were drawn at rest, in the steady state of unloaded pedaling, and regularly throughout the work: at min 2 and every subsequent 4 min for lactate and pyruvate (2 ml); at min 4 and every subsequent 4 min for blood–gas and acid–base variables (1–2 ml); at min 8 and every subsequent 8 min for catecholamines (5–7 ml). PO_2, PCO_2, pH and $[HCO_3^-]$ were measured in duplicate using standard electrodes (Instrumentation Laboratories, #1303) calibrated before and after each measurement with certified standard buffers and gases. Lactate and pyruvate levels ([lac], [pyr]) were measured by standard enzymatic techniques (Olsen 1971); and those of norepinephrine and epinephrine ([NE], [E]) by high-performance liquid chromatography (Watson 1981).

Statistical analysis of measurements was performed using a repeated measures ANOVA. Differences between responses at specific time points were assessed by Tukey's least-significant difference test ($P < 0.05$).

3. Results

A representative example of the P–t relation for high-intensity cycling is presented in figure 1 (upper panel) and, as judged by the excellent linear fit of the transformed P–1/t relation (lower panel), is well described by a hyperbola. The slope (W′) and intercept (P_{LL}) parameters averaged 14·6 kJ and 197 W, respectively (table 2). P_{LL} was substantially greater than $\hat{\Theta}_{lac}$ in all subjects (averaging 164% of $\hat{\Theta}_{lac}$ and 69% of P_{max} obtained on the incremental test: tables 1 and 2) and thus occurred at 46% of the difference between $\hat{\Theta}_{lac}$ and P_{max}. These data were less well described by a mono-exponential model.

For all subjects, the prolonged constant-load test conducted at P_{LL} (i.e. 197 ± 12 W) was continued for the entire 24 min without appreciable duress. In all but one of the $> P_{LL}$ tests (i.e. 213 ± 15 W), subjects fatigued prior to the target duration of 24 min (average duration $17{\cdot}7 \pm 1{\cdot}2$ min).

The temporal characteristics of the $\dot{V}O_2$ responses to the P_{LL} and the $> P_{LL}$ tests are displayed in figure 2 for two subjects (upper panel) and for the entire group (lower panel); these two subjects were chosen to represent examples of the early and more-

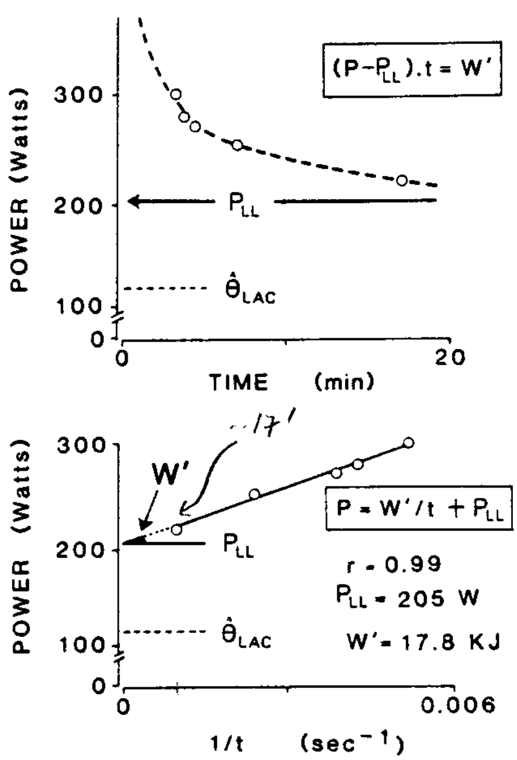

Figure 1. Upper panel: The power–duration (P–t) relationship for high-intensity exercise. Lower panel: Determination of parameters W′ and $> P_{LL}$ from the linear $P - 1/t$ formulation. Subject 1. $\hat{\Theta}_{lac}$ is the estimated lactate threshold.

Table 2. Parameters of the power-duration relationship

Subject	P_{LL} (W)	P_{LL} ($l\,min^{-1}$)	W' (kJ)	r
1	205	2·72	17·8	0·99
2	179	3·02	24·0	0·98
3	222	3·66	12·6	0·97
4	156	2·84	11·5	0·96
5	179	2·64	11·9	0·99
6	171	2·70	14·2	0·98
7	206	3·30	13·2	0·99
8	259	3·07	11·5	0·99
Mean	197	2·99	14·6	0·98
±S.E.	12	0·11	1·6	0·004

P_{LL} (expressed in units of power and $\dot{V}O_2$) and W' represent the intercept and slope parameters, respectively, of the power–duration relationship for high-intensity exercise; r is the correlation coefficient.

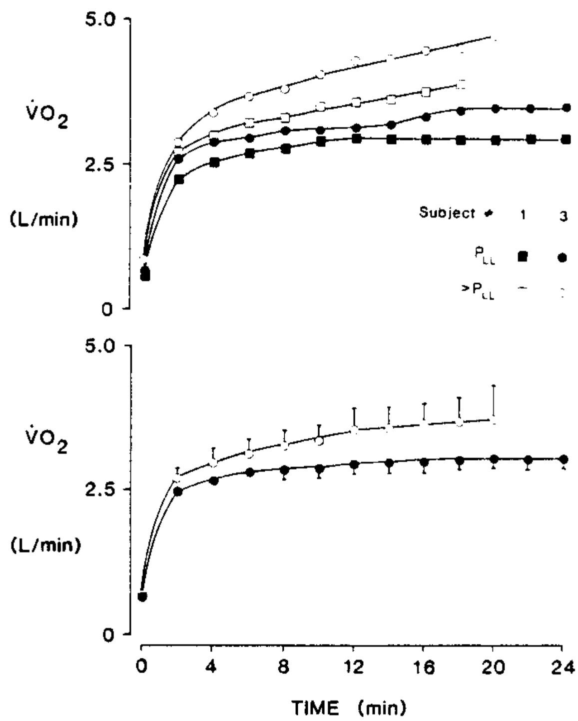

Figure 2. Upper panel: O_2 uptake ($\dot{V}O_2$) responses for two subjects during constant-load exercise at P_{LL} (solid symbols) and $>P_{LL}$ (open symbols). Lower panel: corresponding group mean responses (±s.e.).

delayed attainment of a steady state in $\dot{V}O_2$. For the P_{LL} test, $\dot{V}O_2$ increased rapidly over the first few minutes of the work, rising subsequently at a slower rate to attain a new steady-state of $2{\cdot}99 \pm 0{\cdot}11$ $l\,min^{-1}$ (79·4% ± 3·1% of $\dot{V}O_2$max: table 1) within about 18 min on average, although ranging from about 12 min for subject 1 to almost 20 min for subject 3 (figure 2, upper panel). The P_{LL} $\dot{V}O_2$ profile contrasted with that resulting from the $>P_{LL}$ test, for which no steady state was evident; i.e. a significant group effect on $\dot{V}O_2$ (P_{LL} vs $>P_{LL}$) was demonstrated (repeated measures ANOVA), and there was a significant increase of $\dot{V}O_2$ during the final 4 min of the $>P_{LL}$ test but not over the corresponding interval for the P_{LL} test (i.e. min 16–20 vs. min 20–24, respectively) (Tukey's test). Thus, for the $>P_{LL}$ test $\dot{V}O_2$ rose gradually though systematically to achieve a value at fatigue of $3{\cdot}70 \pm 0{\cdot}38$ l min^{-1} ($97{\cdot}1 \pm 5{\cdot}9$% of $\dot{V}O_2$max: table 1); the end-exercise $\dot{V}O_2$ was not different from $\dot{V}_2$max (t-test) (in three subjects, it exceeded $\dot{V}O_2$max slightly; in the remaining five, it was either equal to or slightly less than $\dot{V}O_2$max).

The $\dot{V}O_2$ attained at the end of the P_{LL} and $>P_{LL}$ tests in each subject was compared with the corresponding $\dot{V}O_2$ predicted from the linear extrapolation of the sub-$\hat{\Theta}_{lac}$ $\dot{V}O_2$–P relation; i.e. predicted $\dot{V}O_2 = \dot{V}O_2(u) + P\,(d\dot{V}O_2/dP)$, where $\dot{V}O_2(u)$ is the $\dot{V}O_2$ for unloaded pedalling and $d\dot{V}O_2/dP$ is the slope of the sub-$\hat{\Theta}_{lac}$ $\dot{V}O_2$–P relation for incremental exercise. The actual $\dot{V}O_2$ was found to exceed the predicted value by an average of 448 ± 95 ml min^{-1} for the P_{LL} test and by 674 ± 235 ml min^{-1} for the $>P_{LL}$ test.

Arterialized venous blood [lac] increased for both the P_{LL} and $>P_{LL}$ tests (figure 3). At P_{LL}, [lac] stabilized at a level of $5{\cdot}6 \pm 0{\cdot}9$ mM l^{-1} in the later stages of the work (i.e. [lac] was not different between min 18 and 22), whereas it rose progressively throughout the $>P_{LL}$ test to attain an average value of $11{\cdot}3 \pm 1{\cdot}4$ mM l^{-1} at fatigue. The profiles of blood [pyr] were essentially similar for both tests (the tendency for [pyr] to rise slightly more rapidly in the $>P_{LL}$ test was not

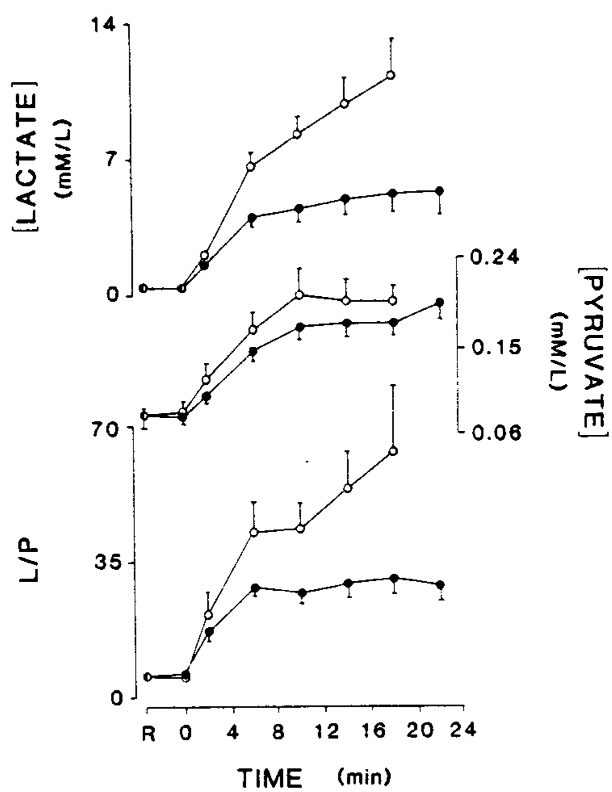

Figure 3. Group mean responses (±s.e.) for [lactate], [pyruvate] and [lactate]/[pyruvate] ratio (L/P) in arterialized venous blood during constant-load exercise at P_{LL} (solid circles) and $>P_{LL}$ (open circles). There was no significant difference between the [lactate] at minutes 18 and 22 for the P_{LL} test ($p > 0{\cdot}05$).

significant). The [lactate]/[pyruvate] ratio, however, was quite stable from ~6 min onwards in the P_{LL} test, but increased systematically throughout the $>P_{LL}$ test.

The [HCO_3^-] response largely mirrored that of [lac] (figure 4), falling continuously and to a greater extent for the $>P_{LL}$ test, while approaching a new stable level within ~20 min for the P_{LL} test. For both tests, arterialized venous pH fell abruptly over the first few minutes of the work (figure 4). It subsequently stabilized for the P_{LL} test (even evidencing a slight rise towards the end of the test), reflecting a fall in arterialized venous PCO_2 that was out of proportion to the fall in [HCO_3^-] (figure 4). In contrast, for the $>P_{LL}$ test, pH continued to decline, though at a slower rate than initially; in this case, the fall in PCO_2 was thus insufficient to constrain the falling pH. In both tests, the fall in the pH preceded that of PCO_2.

$\dot{V}_E$ showed an early marked hyperpnea followed by a more gradual increase for both P_{LL} and $>P_{LL}$ tests (figure 5). In both cases, this reflected an intial rapid increase in V_T, which then plateaued at an average value of ~2·5 l, with frequency generating the subsequent, more slowly developing phase of the hyperpnea (figure 5).

T_{rec} increased progressively throughout the work, by 0·81 ± 0·16°C at the end of the P_{LL} test and by 0·98 ± 0·30°C at the end of the $>P_{LL}$ test (figure 6) from a resting value of 37·25 ± 0·35°C. These end-exercise differences in T_{rec} were not significant. [NE] and [E] also evidenced systematic and progressive increases, which were more marked for the $>P_{LL}$ test (figure 7); the [NE] increase was substantially greater than that of [E].

Finally, the degree of correlation between the $\dot{V}O_2$ profile and the profiles for

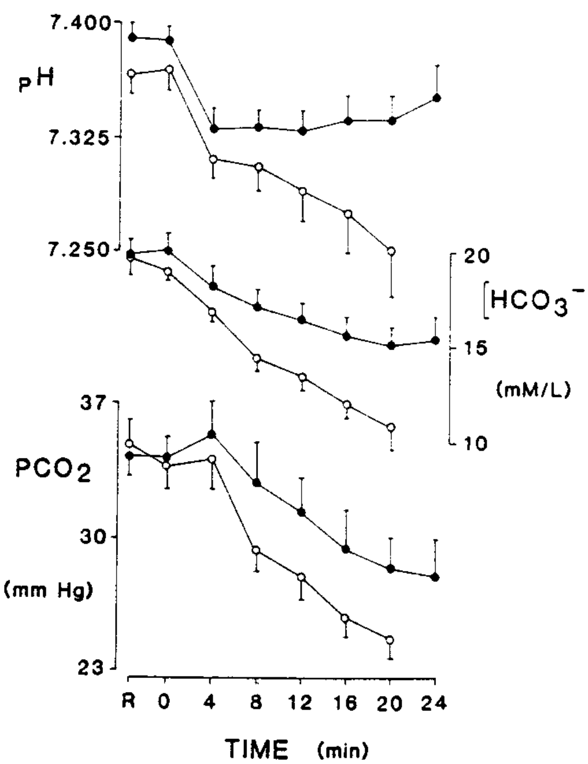

Figure 4. Group mean responses (±s.e.) for pH, bicarbonate [HCO_3^-] and PCO_2 in arterialized venous blood during constant-load exercise at P_{LL} (solid circles) and $>P_{LL}$ (open circles).

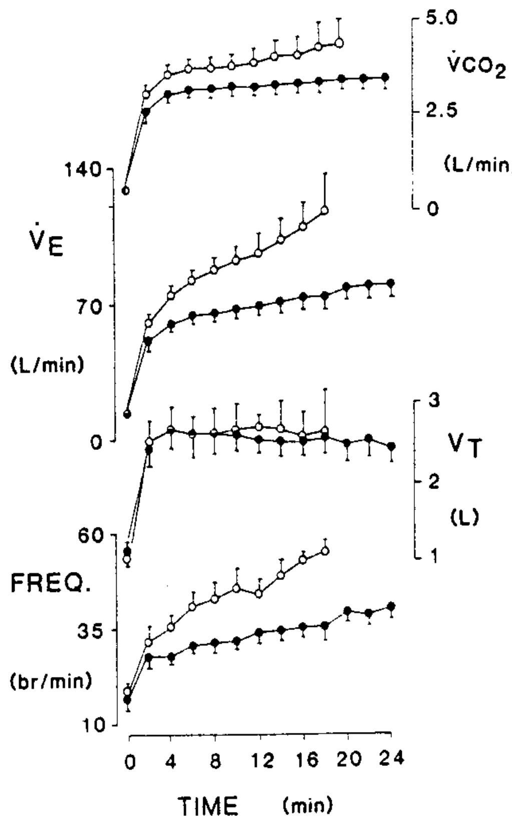

Figure 5. Group mean responses (±s.e.) for CO_2 output ($\dot{V}CO_2$), ventilation ($\dot{V}_E$), tidal volume (V_T) and breathing frequency (breaths/min) during constant-load exercise at P_{LL} (solid circles) and $>P_{LL}$ (open circles).

[lac], [NE], $\dot{V}_E$ and T_{rec} was examined collectively for the P_{LL} and $>P_{LL}$ tests. As it was the delayed, slower phase of the $\dot{V}O_2$ kinetics that was our focus, we deliberately neglected the initial phase of response (min 1–4 for [NE], $\dot{V}O_2$ and T_{rec}; and, because of the timing of the blood sampling, min 1–2 for [lac]); subsequently, responses were expressed as increments occurring over successive 4-min time bins. Figure 8 illustrates

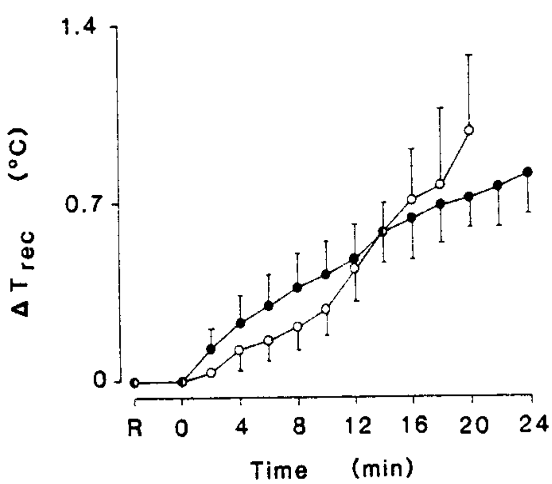

Figure 6. Group mean responses (±s.e.) for the increase in rectal temperature (ΔT_{rec}) during constant-load exercise at P_{LL} (solid circles) and $>P_{LL}$ (open circles).

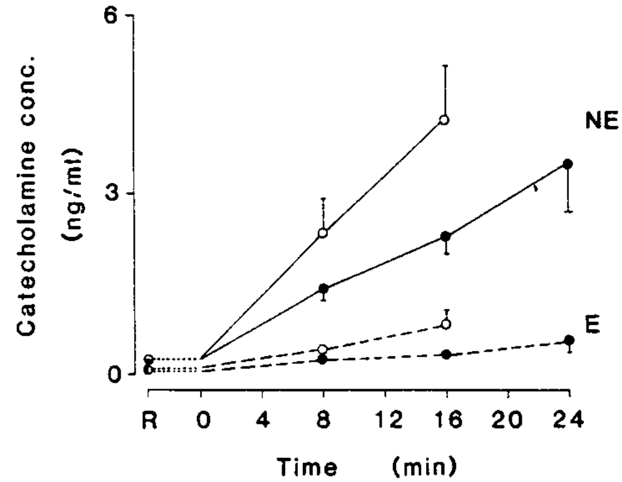

Figure 7. Group mean responses ($\pm$s.e.) for norepinephrine (NE: unbroken lines) and epinephrine (E: broken lines) concentrations in arterialized venous blood during constant-load exercise at P_{LL} (solid circles) $> P_{LL}$ (open circles).

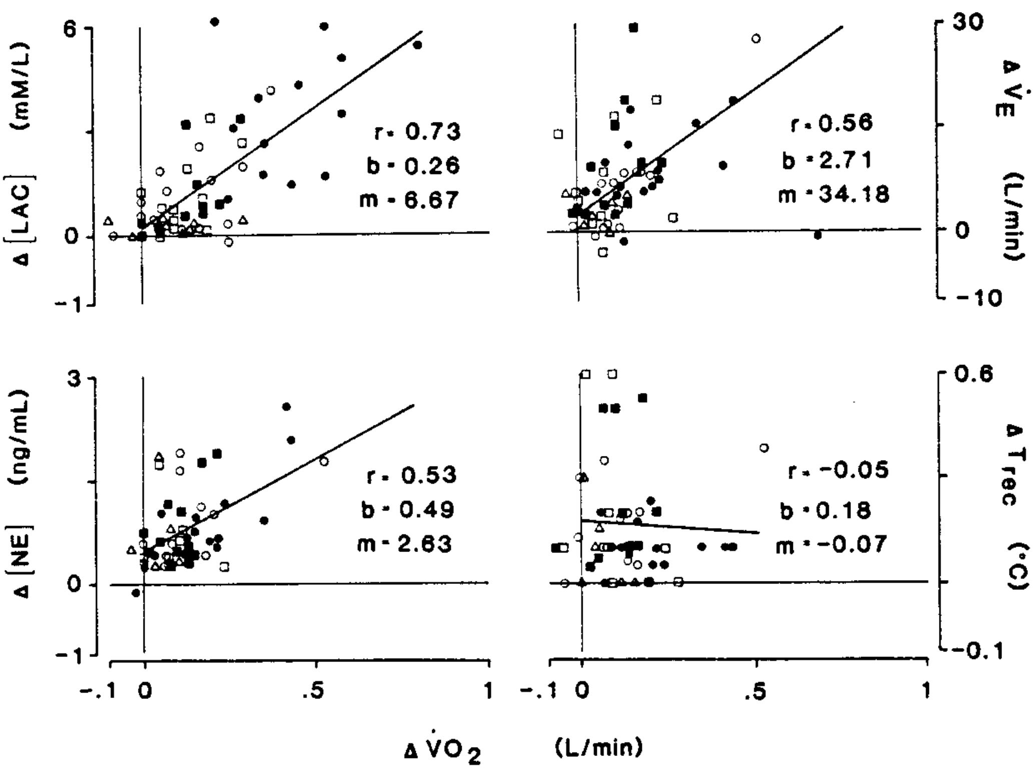

Figure 8. Correlation between temporal responses of lactate concentration (Δ[lac]), ventilation ($\Delta\dot{V}_E$), norepinephrine (Δ[NE]) and rectal temperature (ΔT_{rec}) with the corresponding response of O_2 uptake ($\Delta\dot{V}O_2$), subject-by-subject, throughout the P_{LL} tests and the $> P_{LL}$ tests. Data for each subject and each test type are averaged into successive 4-min bins (●, ○, ■, □, △), starting from the 4th min of the work for $\dot{V}_E$, [NE] and T_{rec} and, owing to the timing of the blood sampling, from the 2nd min for [Lac]. Unbroken lines are best-fit linear regressions; *m* and *b* are the slope and intercept, respectively, and *r* is the correlation coefficient. See the text for further details.

that the $\dot{V}O_2$ response ($\Delta\dot{V}O_2$) correlated most strikingly with that of lactate (Δ[lac]), which is consistent with the $\dot{V}O_2$ and [lac] responses eventually stabilizing for the P_{LL} test while continuing to rise throughout the $>P_{LL}$ test (figures 2 and 3). Although there were positive correlations between the $\dot{V}O_2$ response and both those of [NE] and $\dot{V}_E$ (figure 8), these were considerably less prominent than for [lac]; i.e. while [NE] and $\dot{V}_E$ continued to rise throughout both the P_{LL} and the $>P_{LL}$ tests (figures 5 and 7), this was the case for $\dot{V}O_2$ only for the $>P_{LL}$ test (figure 2). A similar situation accounted for the poor correlation between T_{rec} and $\dot{V}O_2$ (figures 6 and 8). The results of this correlational analysis were reinforced by the repeated measures ANOVA: a significant time effect was evident on $\dot{V}O_2$, $\dot{V}_E$, [lac], [NE] and T_{rec}; however, only $\dot{V}O_2$, $\dot{V}_E$ and [lac] yielded a between-group difference.

4. Discussion

Based upon the empirically demonstrable hyperbolic P–t relation for high-intensity exercise (Monod and Scherrer 1965, Moritani *et al.* 1981, Hughson *et al.* 1984), supra-Θ_{lac} work may be considered to encompass two intensity domains: (i) between Θ_{lac} and P_{LL}, $\dot{V}O_2$ can attain a new steady state and hence allow the performance of prolonged exercise; (ii) $>P_{LL}$, fatigue ensues more rapidly (as a hyperbolic function of P), with $\dot{V}O_2$ eventually reaching $\dot{V}O_2$max (as determined from incremental testing) and thus, in this region, $\dot{V}O_2$ behaves as a function of both time *and* power. P_{LL} therefore represents a demarcation between 'heavy-intensity' exercise (which may be prolonged despite a sustained metabolic acidosis) and 'severe-intensity' exercise (leading to progressive and inexorable increases both of $\dot{V}O_2$ to its maximum and of the degree of metabolic acidosis). Thus, P_{LL} also represents a discontinuity in the *achievable* $\dot{V}O_2$–P relation, as is readily apparent from the $>P_{LL}$ tests for which the power was appreciably lower than P_{max} on the incremental test (figure 2; table 1). Hence, care must be taken when extrapolating the results of an incremental exercise test to sustained constant-load work, in that a $\dot{V}O_2$max will be manifest at power outputs well below those achieved on (or predicted from) the incremental test.

Previous investigators have demonstrated that supra-Θ_{lac} $\dot{V}O_2$ kinetics evidence a slowly rising, delayed phase superimposed on the rapid, exponential phase characteristic of sub-Θ_{lac} work, and which may proceed to a new steady state unless fatigue ensues (Whipp and Wasserman 1972, 1986, Linnarsson 1974, Hagberg *et al.* 1978, Whipp and Mahler 1980). The asymptotic value attained by $\dot{V}O_2$ at these high power outputs represents an increase in $\dot{V}O_2$ *beyond*, rather than *towards*, the 'steady-state' $\dot{V}O_2$ expected from a linear extrapolation of the sub-Θ_{lac} $\dot{V}O_2$–P relation (Whipp and Mahler 1980, Whipp and Wasserman 1986). Our results indicate that this occurs not only in $>P_{LL}$ tests, but also in the P_{LL} tests (for which $\dot{V}O_2$ eventually stabilized: figure 2). This extra O_2 appears to be the result of some metabolic process which is not evident during constant-load work $<\Theta_{lac}$ (Whipp and Wasserman 1972). Whatever the causative factors may be, their influence appears eventually to be stabilized if the power is constrained to lie at or $<P_{LL}$.

While blood PCO_2 fell in both P_{LL} and $>P_{LL}$ tests (presumably reflecting compensatory hyperventilation for the metabolic acidosis), this was delayed relative to the pH fall (figure 4). This delay could indicate a mild hyperventilation prior to the work (owing to apprehension or nervousness in some subjects) serving to mask hypocapnia at the onset of the acidosis. Alternatively, the $\dot{V}_E$ kinetics in response to the acidosis may be slow: delayed hypocapnia is observed $>\Theta_{lac}$ with rapidly incrementing exercise tests (increment duration $\leqslant 1$ min) (i.e. an 'isocapnic buffering' phase) but not with more slowly incrementing profiles (>4 min) (Wasserman *et al.* 1977); this might reflect a time-related or pH-related threshold for carotid body activation (Whipp 1983), coupled with slow wash-out kinetics of CO_2 from body stores (Nunn 1977).

Empirical support for the P–t relation for high-intensity exercise being hyperbolic is provided by the excellent fit of our data (figure 1), and those of others (Monod and Scherrer 1965, Moritani *et al.* 1981, Hughson *et al.* 1984), to a hyperbola. Furthermore, with the reasonable assumption that some fatigue-inducing factor (intramuscular pH; available concentrations of high-energy phosphates) decreases monotonically with time (e.g. linearly or exponentially) towards some low, limiting value during constant-load work, then the parameter characterizing this rate of decline (i.e. the slope or the time constant, respectively) should be a function of P; a necessary consequence would be that the time to reach this low limiting value should decrease hyperbolically as P increases. In this context, the shortcomings associated with the exponential characterization of the P–t relation proposed by Wilkie (1981) should be recognized; i.e. $P = E + (A/t) - E\tau\,(1 - e^{-t/\tau})/t$ (eqn. 3), where A is the anaerobic energy store, τ the $\dot{V}O_2$ time constant, and E the power equivalent of $\dot{V}O_2$max. Thus, (i) the suggested τ (10 s) is much too short for man (Linnarsson 1974, Whipp and Wasserman 1972, Whipp and Mahler 1980); (ii) $\dot{V}O_2$ kinetics at this intensity are not monoexponential (Whipp and Wasserman 1972, 1986, Linnarsson 1974, Hagberg *et al.* 1978, Whipp and Mahler 1980) and (iii) as all power outputs $> P_{LL}$ eventually lead to the attainment of $\dot{V}O_2$max (figure 2), it is difficult to envisage how E can reasonably be regarded as a fundamental parameter of this relation. Interestingly, however, eqn. 3 actually reduces to a hyperbola (eqn. 2; Moritani *et al.* 1981) over the range of fatiguing work durations which we found in our study: as $t \gg \tau$, the exponential term in eqn. 3 becomes disappearingly small: $P = E + A/t$; and we found Wilkie's data to yield an excellent linear fit to the hyperbolic ($P - 1/t$) transform.

Lactate and its related H^+ ion appear to be the most likely candidates underlying the power asymptote of the $P - t$ relation and the associated discontinuity in the $\dot{V}O_2$–P relation. In our study, the blood [lac] profile both at and $> P_{LL}$ was highly correlated with that of $\dot{V}O_2$ (figures 2, 3 and 8). Also, an increased blood [lactate] can augment metabolic rate, possibly through an increased rate of glycogen formation from lactate in the liver and, perhaps, in skeletal muscle (Alpert and Root 1954, Bertram *et al.* 1967, Haggendal 1971, Ryan *et al.* 1979) The mechanisms of this effect are not clear: an involvement of hepatic gluconeogenesis has been suggested in dog (Barnard *et al.* 1970), but considerations of hepatic perfusion during heavy exercise in man (Rowell 1971) argue against a significant contribution from this source. Conversion of lactate to glycogen in skeletal muscle has also been proposed (Bendall and Taylor 1970, McLane and Holloszy 1979), although it is not clear which fibre population may be responsible, as (i) slow-twitch (ST) fibres seem to have a low glyconeogenic capacity, at least in rat (McClane and Holloszy 1979); (ii) during prolonged exercise, fast-twitch (FT) fibres may not become glycogen-depleted until near exhaustion; and (iii) thermodynamic considerations argue against high rates of glycolysis and glyconeogenesis occurring simultaneously in the same fibre.

Epinephrine and norepinephrine can influence $\dot{V}O_2$ through augmented liver and skeletal muscle glycogenolysis and adipose tissue and skeletal muscle lipolysis (Gollnick 1978). However, our data at P_{LL} show $\dot{V}O_2$ to have stabilized while blood [E] and [NE] were still rising (figures 2 and 7); and despite $\dot{V}O_2$, [E] and [NE] continuing to rise throughout the $> P_{LL}$ test (figures 2 and 7), the [NE]–$\dot{V}O_2$

correlation was low (figure 8). Thus, the involvement of catecholamines in the augmented $\dot{V}O_2$ response at and $> P_{LL}$ is likely to be minor. Most investigators agree that increases in core temperature during exercise are unlikely to affect muscle metabolism significantly (Dill *et al.* 1931, Ekblom *et al.* 1971, Rowell 1971; cf. Hagberg *et al.* 1978). Our data are consistent with this view†: there was no demonstrable association between $\dot{V}O_2$ and T_{rec} (figures 6 and 8).

A serial, progressive recruitment of proportionally more FT fibres in the contracting muscles as the work progressed could potentially augment muscle O_2 consumption and thus generate a significant slow component in the $\dot{V}O_2$ response profile. Prolonged heavy exercise can induce a preferential utilization of ST and FT oxidative fibres, with a secondary recruitment of FT oxidative/glycolytic and FT glycolytic fibres as the fibres initially recruited become glycogen-depleted (Vollestad *et al.* 1984); and there is evidence to suggest that the energy cost per unit force output is higher for FT fibres than for ST fibres (Crow and Kushmerick 1982). To our knowledge, however, the implications of these findings have not been rigorously addressed in man.

It is unlikely that the O_2 cost of breathing contributed significantly to the augmented $\dot{V}O_2$ response $> P_{LL}$. Recognizing that accurate estimation of the O_2 cost of breathing is difficult (McKerrow and Otis 1956), only minor increases in $\dot{V}O_2$ attend the substantial hyperpnea of prolonged, high-intensity work (Ekblom *et al.* 1971, Rowell 1971; figures 2 and 5; cf. Hagberg *et al.* 1978). And despite the demonstration of a significant (but low) correlation between $\dot{V}_E$ and $\dot{V}O_2$ for both the P_{LL} and $> P_{LL}$ tests (figure 8), there were individual cases in which $\dot{V}_E$ was clearly rising at P_{LL} when $\dot{V}O_2$ had stabilized. Likewise, although cardiac output is high beyond P_{LL}, any influence on $\dot{V}O_2$ from the O_2 cost of cardiac work is likely to be trivially small (Dill *et al.* 1931). An acidotic state in the working muscle may plausibly decrease contraction efficiency; however, $\dot{V}O_2$ appears to be unaffected by a pronounced systemic acidosis or alkalosis during cycling (Jones *et al.* 1977). The mechanical efficiency of heavy cycling may be decreased because of additional work performed by the upper body: however, the slowly rising phase of $\dot{V}O_2$ can still be discerned in the absence of exaggerated movements of the upper body and/or pulling on the hand bars. Finally, while the thyroid hormones T_3 and T_4 stimulate metabolism and thus have the potential to augment $\dot{V}O_2$, not only are their levels little changed during prolonged exhausting exercise (Galbo *et al.* 1977), but also patients with thyrotoxicosis have been shown to have a normal work efficiency (Bishop *et al.* 1955).

We therefore conclude that during constant-load exercise tests performed at power outputs above the power asymptote (P_{LL}) of the P–t relationship, $\dot{V}O_2$ will continue to increase throughout the test eventually attaining a maximum $\dot{V}O_2$ and the limit of work tolerance. Although numerous factors are likely to contribute to this slow, continued $\dot{V}O_2$ increase, mechanisms related to the sites and routes of lactate metabolism appear to be dominant mediators. Consequently, because of the inability to stabilize $\dot{V}O_2$ for the supra-P_{LL} power outputs, it may not be possible to justify a 'required' $\dot{V}O_2$ within this intensity domain.

† Recognizing that the thermolabile or 'Q_{10}' effect on $\dot{V}O_2$ will be influenced predominantly by the tissues generating the metabolic responses, it is important to recognize that although the temperature in the exercising muscle is likely to rise to a greater extent than core temperature, their rates of change are similar for the exercise intensities and durations utilized in this investigation (Saltin *et al.* 1968). The percentage thermolabile increase in $\dot{V}O_2$ from this mechanism ($= Q_{10}^{(\Delta t/10)}$) yields a predicted 20% increase for an increase of temperature (Δt) from 37 to 39°C, assuming a Q_{10} of 2·5.

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Pour un exercise intense sur ergocyle, la puissance (P) peut être correctement décrite comme étant une fonction hyperbolique de la durée (t) de travail tolérable: $P=(W'/t)+P_{LL}$ où W′ est une constante et P_{LL} la limite inférieure (asymptote) pour laquelle P est produite avec une consommation d'oxygène ($\dot{V}O_2$) situé au-dessus du seuil estimé des accroissements ($\hat{\Theta}_{lac}$) du lactate circulant, mais en-dessous du $\dot{V}O_2$max atteint pendant un exercice incrémenté. Cette relation suggère qu'au-dessus de P_{LL}, seulement une certaine quantité de travail (W′) peut être èeffeçtuée quelle que soit sa cadence, avec une $\dot{V}O_2$max atteinte lors de la fatigue. Donc P_{LL} définit un point de rupture dans la relation $\dot{V}O_2$–P pour un exercice supra-$\hat{\Theta}_{lac}$. Afin de déterminer quels sont les facteurs responsables de l'accroissement continu de la $\dot{V}O_2$ (jusqu'à la valeur de fatigue maximum) pour des rendements de puissance supérieurs à P_{LL}, nous avons examiné l'évolution chronologique des réponses métaboliques (température rectale, lactate, pyruvate, noradrénaline et adrénaline plasmatiques) et respiratoires ($\dot{V}_E$, $\dot{V}O_2$, $\dot{V}CO_2$; le pH, PCO_2 et HCO_3^- sanguins) de 8 sujets masculins lors d'un exercice à charge constante sur ergocyle au niveau de la P_{LL} (jusqu'à l'épuisement, c.à.d. >24 min.). La $\dot{V}O_2$ a présenté un état stable retardé pour la P_{LL}, en dépit de la poursuite de l'accroissement des taux de catécholamines et de la température centrale. Toutefois le pH et le lactate avaient atteint un plateau. En revanche, la $\dot{V}O_2$ continuait à augmenter lentement pendant toute la durée de l'exercice P_{LL} pour atteindre la $\dot{V}O_2$max. Ce modèle de réponse aux niveaux P_{LL} et $>P_{LL}$ suggère que la phase lente de la réponse $\dot{V}O_2$ corrèle bien avec le profil chronologique du lactate sanguin. Par conséquent, il est possible que le site et le cours du métabolisme de cette variable jouent un rôle déterminant dans la cinétique de la $\dot{V}O_2$ pour les exercices intenses.